Cost-Effective Dispatch Using Grey Wolf Optimization Algorithm: Solution with Diverse Load Pattern

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Abstract -This paper ascertains a new solution to the economic load dispatch (ELD) problem using a grey wolf optimization (GWO) algorithm with diverse load conditions. ELD has formulated as a constrained minimization problem that satisfies certain practical and realistic constraints. In this study, both quadratic and non-convex fuel cost characteristics have considered as an objective function under static and dynamic load environments. The grey wolf optimization algorithm (GWO) has adopted as an optimization tool. It carries the exploration and exploitation process in search space through encircling, hunting, and attacking, and the algorithm has incorporated into a solution repair strategy. Therefore, a new optimum generation schedule has arrived. For demonstration, two test systems have taken with from earlier reports with full consideration, such as transmission loss and valve point loading effect. The simulated results have contrasted with the results attained by previous methods, and the validation reveals that the proposed method is a hopeful optimization tool to find generation schedules at affordable prices.

Keywords— Cost-effective dispatch (CED), Metaheuristic optimization, Grey wolf optimizer (GWO), quadratic and non-convex model

I. PROLOGUE

The growing energy sector is influencing the sustainable development of a nation greatly since rapidly increasing economic development increases energy consumption. Though it includes generation from non-conventional energy resources coal, is the foremost energy resource. In this situation, a suitable forecasting procedure has mandated to balance the production and load demand which is nothing but economic load dispatch. It is a nonlinear optimization problem intent to minimize the operating cost of power generation subjected to a different system and operating constraints. As the fuel cost is expressed higher-order nonlinearity and discontinues, a suitable optimization technique is required.

Conveniently, bat algorithm (BA) [1], artificial neural network (ANN) [2], novel consensus-based distributed control algorithm [3], a hybrid PSO and the artificial bee colony (PSO-ABC) [4],teaching-learning based optimization (TLBO) and Jaya algorithms [5], harmony search (HS) algorithm [6], parallel hurricane optimization algorithm [7] and chaotic improved harmony search algorithm (CIHS) [8] methods have demonstrated static economic dispatch problem in the aspects of solution quality, stable convergence and computation efficiency. The adaptive charged system search algorithm (ACSS) improves the initialization stage inspired by opposite based learning (OBL) and sub-spacing methods [9]. A modified crow search algorithm (MCSA) improves search capability and solution quality with more excellent performance [10].

At the same time, to solve the dynamic economic dispatch (DED) problem, Basu proposes a non-dominated sorting genetic algorithm-II (NSGA-II) which provides the solution in a wide spread and converge near-global solution [11]. Improved bacterial foraging algorithm used a parameter automation strategy and crossover operation in micro BFA to enhance computational efficiency [12]. Elaiw has integrated sequential quadratic programming (SQP) with differential evolution (DE) and particle swarm optimization (PSO) separately to develop hybrid DE-SQP and PSO-SQP algorithms for finding optimal solution [13]. A new modified non-dominated sorting genetic Algorithm-II guarantees the diversity of the population under the premise of the elitism of the residents [14].

In the hybrid chemical reaction optimization (CRO) algorithm, DE has integrated to increase the convergence speed and to provide an optimal global solution [15] whereas, the evolutionary operator of differential evolution (DE) is modified adaptively to avoid the premature (MA-DE) [16] while solving DED. Zongyan proposed a new enhanced harmony search (NEHS) algorithm to solve the DED problem and also beneficial for conducting searching potential and global exploration DED solutions in the solution space [17]. Modified competitive swarm optimizer (MCSO) embroils best global position for updating the particles by introducing a new tri-competition mechanism [18].

By inspiring the superior convergence characteristics of a grey wolf optimizer (GWO) [19], it is exercised in this paper to discover a cost-effective dispatch (CED) under static and dynamic loading environment.

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II. PROBLEM FORMULATION

The cost-effective dispatch problem is intended to minimize the operating cost of the thermal power system under the diverse load demand as follows.

A. Cost-Effective Objective function (OF)

i. Static Load Pattern

$$Minimize\ F = \sum_{i=1}^{N_s}\left[f_{i,k}\left(P_{si,k}\right)\right] \tag{1}$$

ii. Dynamic Load Pattern

$$Minimize\ F = \sum_{k=1}^{T}\sum_{i=1}^{N_s} t_k\left[f_{i,k}\left(P_{si,k}\right)\right] \tag{2}$$

Quadratic cost model:

$$f_{i,k}\left(P_{si,k}\right) = a_i + b_i P_{si,k} + c_i P_{si,k}^2 \tag{3}$$

Non-convex cost model:

$$f_{i,k}\left(P_{si,k}\right) = a_i + b_i P_{si,k} + c_i P_{si,k}^2 + \left| d_i \sin\left\{ g_i\left(P_{si}^{\min} - P_{si,k}\right)\right\}\right| \tag{4}$$

Where,

N_s -Number of units

F -Total generation cost ($)

f_i -Fuel cost of ith thermal unit (TU) ($/hr)

a_i, b_i, c_i -Fuel cost coefficients of ith unit

d_i, g_i -Coefficient of valve-point loading of ith unit

P_{si} -Generation of ith unit

$P_{si}^{\min}$, $P_{si}^{\max}$ -Lower and upper generation limit of ith unit

T -Total number of scheduling period (hr)

k -Sub-interval (hr)

t_k -Sub-interval duration (hr)

B. System constraints

i. Equality Constraints

The Power balance equation is articulated as follows [11]:

$$\sum_{i=1}^{N_s} P_{s,ik} - P_{D,k} - P_{L,k} = 0;\ k \in T \tag{5}$$

Where,

$$P_{L,k} = \sum_{m=1}^{N_s}\sum_{n=1}^{N_s} P_{m,k} B_{mn} P_{n,k} + \sum_{m=1}^{N_s} B_{m,0} P_{m,k} + B_{00}\ ;k \in T \tag{6}$$

P_D -Total power demand (MW)

P_L -Total network loss (MW)

B_{mn}, B_{m0}, B_{00} -Loss coefficients

$P_{m,k}$, $P_{n,k}$ -Power production of the mth and n^{th} index of plants

ii. Inequality Constraints

Generation limits are prescribed as develops [11]:

$$P_{si}^{\min} \le P_{si,k} \le P_{si}^{\max} \qquad i = 1,2,....N_s \tag{7}$$

iii. Practical limitations

Prohibited operating zones (POZ) [9]:

$$\begin{cases} P_{sj}^{\min} \le P_{sj,k} \le P_{sj,1}^{L} \\ P_{sj,m-1}^{U} \le P_{sj,k} \le P_{sj,m}^{L}\ ;m = 2,3,..,ND_j \\ P_{sj,m}^{U} \le P_{sj,k} \le P_{sj}^{\max}\ \ ;m = ND_j \end{cases} \tag{8}$$

Where,

P_{sj}^{L} -Lower limit of j[th] POZ

P_{sj}^{U} -Upmost limit of j[th]POZ

ND -Number of the POZ

iv. Ramp rate limits[9]:

$$\begin{cases} P_{si,k} - P_{si,k-1} \le UR_i\ ;if\ generation\ \ raises \\ P_{si,k-1} - P_{si,k} \le DR_i\ ;if\ generation\ lowers \end{cases} \tag{9}$$

Where,

UR_i -Upper ramp bound of the i^{th}TU

DR_i -Down ramp bound of the ith TU

III. GWO BASED COST-EFFECTIVE DISPATCH

According to Seyedali Mirjalili [18], an influential member called alpha wolf (α) dominates entire colony consisting omega (ω) wolfs with the association of immediate strengthen wolf's beta (β) and delta (δ). Generally, the wolves follow encircling, hunting and attacking mechanisms while foraging the prey. To apply GWO for solving CED, first, the wolf's behavior can be mathematically expressed in terms of a system parameter [19-20]. The following procedure explains the applicability of GWO.

Step-1: As the fuel cost is the function of real power generation, the population of the wolf is actuated about real power to be generated, and it is randomly created in the search space using (10).

$$P_{g,i} = rand * \left(P_{gi}^{\max} - P_{gi}^{\min}\right) + P_{gi}^{\min} \tag{10}$$

The preliminary control variables are represented as matrix form in (11).

$$X = \begin{bmatrix} P_{g1}^{1} & P_{g2}^{1} & . & . & P_{gi}^{1} & P_{gN}^{1} \\ P_{g1}^{2} & P_{g2}^{2} & . & . & P_{gi}^{2} & P_{gN}^{2} \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ P_{g1}^{SP} & P_{g2}^{SP} & . & . & P_{gi}^{SP} & P_{gN}^{SP} \end{bmatrix} \tag{11}$$

It is positioned as like (12) to facilitate the computational process.

$$X^{o} = \left[P_{g1}^{1} \cdot - \cdot P_{g1}^{SP}\ P_{g2}^{1} \cdot - \cdot P_{g2}^{SP}\ ....P_{gi}^{1} \cdot - \cdot P_{gi}^{SP} ... P_{gN}^{1} \cdot - \cdot P_{gN}^{SP}\right] \tag{12}$$

Step-2: The objective function is amended as similar (13) to satisfy the equality constraint. Here λ is a significant numerical multiplier.

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$$AOF = \left(OF + \lambda * \left| \sum_{i=1}^{N} P_{gi} - \left(P_D + P_L \right) \right| \right) \tag{13}$$

Step-3: The robustness of ameliorated objective function is computed using (14) and arranged all populations shown in (12) as per the fitness value from best to worst. The first best agent is noted as alpha, and the immediate two best agents are considered as beta and delta wolves correspondingly.

$$\text{Fitness} = \text{AOF} \tag{14}$$

Step-4: Concerning the alpha, beta, and delta's location rest of the agent (omega) position to be updated using (15).The state of apiece wolf mimics an impending solution that included real power procreation of CED problem. The latest location of every wolf may dismiss acceptable degrees and it is restricted in that span.

$$\vec{X}(t+1) = \frac{\vec{X}_1(t) + \vec{X}_2(t) + \vec{X}_3(t)}{3} \tag{15}$$

Where,

$$\begin{aligned} \vec{X}_1 &= \vec{X}_{aplha} - \vec{A}_1 \cdot \left(\vec{D}_{alpha} \right) \\ \vec{X}_2 &= \vec{X}_{beta} - \vec{A}_2 \cdot \left(\vec{D}_{beta} \right) \\ \vec{X}_3 &= \vec{X}_{delta} - \vec{A}_3 \cdot \left(\vec{D}_{delta} \right) \end{aligned} \tag{16}$$

$$\begin{aligned} \vec{D}_{alpha} &= \left| \vec{C}_1 \cdot \vec{X}_{alpha} - \vec{X} \right| \\ \vec{D}_{beta} &= \left| \vec{C}_2 \cdot \vec{X}_{beta} - \vec{X} \right| \\ \vec{D}_{delta} &= \left| \vec{C}_3 \cdot \vec{X}_{delta} - \vec{X} \right| \end{aligned} \tag{17}$$

Step-5: Perform step-2 for updated agents to identify a globally optimal solution.

Step-6: Adjust the generation of dth entity based on solution fix procedure by (18).

$$\begin{aligned} B_{dd} P_{g\,d}^2 + \left(2 \sum_{m=1}^{(N-1)} B_{d,m} P_m - 1 \right) P_{g\,d} + \left(\sum_{m=1}^{(N-1)} \sum_{n}^{(N-1)} P_m B_{mn} P_n \right. \\ \left. + \sum_{m=1}^{(N-1)} B_{m,0} P_m - \sum_{\substack{m=1 \\ m \neq d}}^{(N-1)} P_m + B_{00} + P_D \right) = 0 \end{aligned} \tag{18}$$

Step-7: The best generation schedule is to be satisfied not only with its inequality constraints but also meets POZ and ramp rate limits by (19) and (20), respectively.

$$P_i = \begin{cases} P_{i,l}^L & rand \leq 0.5 \\ P_{i,k}^U & rand > 0.5 \end{cases} \quad k = 2,3,..N_i \tag{19}$$

$$\left.\begin{aligned} P_i^{r\max} &= \min\left\{ P_i^{m\max}, \left(P_i^o + UR_i \right) \right\} \\ P_i^{r\min} &= \max\left\{ P_i^{\min}, \left(P_i^o - DR_i \right) \right\} \\ P_i^{r\min} &\leq P_i \leq P_i^{r\max} \end{aligned}\right\} \tag{20}$$

Step-8: If the iteration is less than the maximum number of a cycle, depart to step 2. Alternatively, the GWO breaks off.

IV. SIMULATION RESULTS AND DISCUSSION

The GWO is programmed in MATLAB 8.1 environment and simulated on the Intel Core i5 processor personal computer. Two test systems (TS) are considered for this study, and the first system embodies10 generating units encompass quadratic and non-convex cost characteristics. The load demand is set into 2000MW, and network loss is let into a goal. Additionally, the cost curve and B-coefficients are referred to from [6]. In the second test system, the unit records of TS-I have customized and the load demand was scheduled 24 hours [11]. The numerical results have demonstrated into two case studies; one is static dispatch whereas, the second is dynamic dispatch.

A. Static dispatch

The applicability of the GWO algorithm in minimizing fuel cost is tested with a medium-size thermal power system (TPS), which has ten units. The intended algorithm is converged at a minimum operating cost of $ 111140.8317 and 111355.0265 per hour for quadratic and non-convex cost model TPS respectively. Figure 1 portrays fast and steady convergence characteristics of GWO while minimizing fuel costs. The best total dispatch attained by the GWO, fuel cost is presented in Table I. The GWO has settled with the new least-cost dispatch.

Table I Cost-Effective Dispatch of TS-I

Generation (MW)	Demand 2000 MW	
	Quadratic cost model	Non-convex cost model
P1	55.0000	52.8003
P2	76.4728	80.0000
P3	115.7513	100.1980
P4	95.0782	95.8466
P5	98.4812	93.5323
P6	102.6063	115.0453
P7	281.5338	263.4976
P8	335.5135	338.8421
P9	451.0658	470.0000
P10	467.4685	469.2508
Total	2078.9714	2079.0130
Losses (MW)	78.9714	79.0130
Iterations	500	500
Fuel Cost ($/hr)	**111140.8317**	**111355.0265**

Table II Comparison of cost-effective solution with other methods for TS-I

Methods	Fuel cost ($/hr)
CIHSA [8]	111497.6310
ABCPSO [4]	111500.0000
DE [4]	111500.0000
PHOA [7]	112130.8500
GWO	**111355.0265**

Further, the best cost attained by the GWO for non-smooth price modeled TPS has compared with recent reports in Table II. The attained total fuel cost using GWO is the least when compared with the earlier reports. Moreover, the saving in fuel price that GWO has received is portrayed in Figure 2, which shows the solution quality improvement concerning other contestant algorithms.

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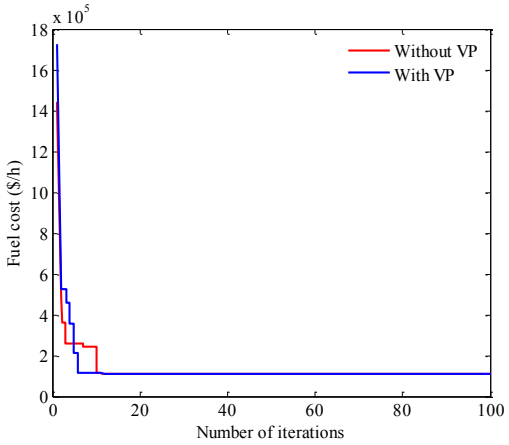


Fig. 1 Convergence characteristics of GWO for TS-I

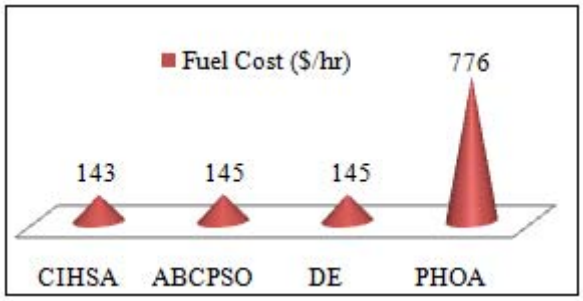


Fig.2 Fuel cost improvement obtained by GWO for unit TS-I

B. Dynamic dispatch

In real-time operations, the demands are time-varying; thus, the TPS operation has become more complicated in a dynamic environment. The applicability of the suggested algorithm is verified for the standard test system. It consists of ten generation units with quadratic and non-smooth fuel cost functions, and other operational constraints such as ramp rate limits and POZ are considered. The demand for the system was divided into 24 intervals. The GWO is executed for 100 iterations and its convergence curve is depicted in Figure 3. From the characteristics, it can be observed that the algorithm is converged at minimum objective value quickly for both quadratic and non-convex cost models.

In Table III, total generation and loss corresponding to minimum fuel cost for TS-II without and with valve point loading effect are comprehended; it is understood that the fuel cost is increased slightly due to the valve point loading effect. While optimizing TPS under dynamic load pattern, its best solution is distinguished with the earlier methods in Table IV, to validate the effectiveness of GWO and it reveals that GWO outperforms the contestant's algorithm. Additionally, the potential in saving fuel cost in ($) is pictured in Figure 4, and it shows that GWO saves fuel costs in a tremendous amount concerning all methods.

Table III Comparison of feasible solution of GWO for TS-II

Attributes	Without Valve Point	With Valve Point
Total Generation (MW)	41062.3270	41065.6206
Total Losses (MW)	954.3270	957.6206
Total Fuel Cost ($)	**2444072.4953**	**2462593.6473**

Table IV Comparison of cost-effective solution with other methods for TS-II

Methods	Fuel cost ($)
CRO [15]	2481613.3800
HCRO [15]	2479931.3800
DE-SQP [13]	2465910.8369
PSO-SQP [13]	2466805.7427
MA-DE [14]	2492451.0000
IBFA [12]	2481733.2570
RCGA [11]	2516800.0000
MRGA [16]	2497000.0000
NSGA-II [11]	2516800.0000
NEHS [17]	2463500.8405
GWO	**2462593.6473**

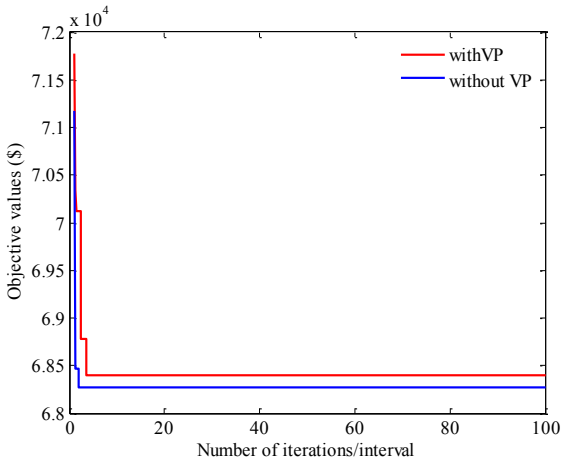


Fig. 3 Convergence characteristics of GWO for TS-II

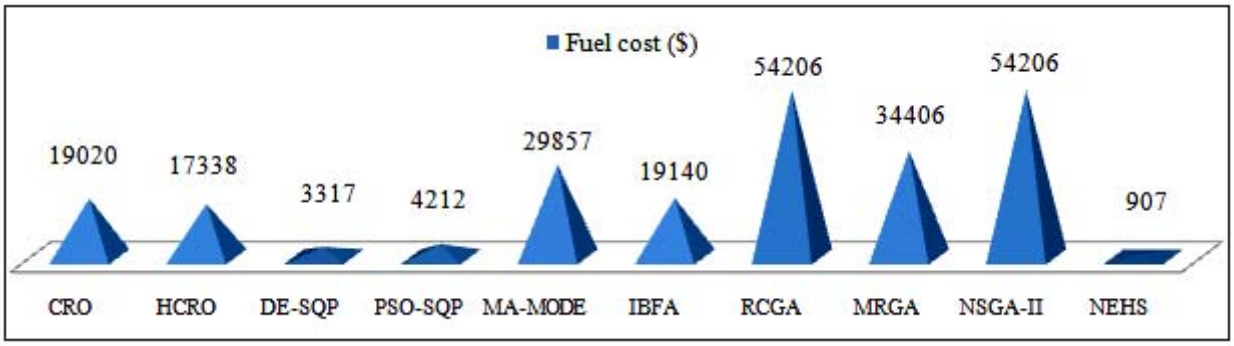


Fig. 4 Fuel cost improvement obtained by GWO with other methods for TS-II

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V. CONCLUSION

It is summarized that cost-effective dispatch is framed as a nonlinear optimization problem. In which the objective function is the fuel cost of TPS that included valve point loading effect, and modeled both static and dynamic load environment. The constraints are the power balance equation, and minimum and maximum power generation limits. For realistic operation, POZ and up/down ramp rate limits have considered in the problem formulation. Proficient constraints handling mechanisms are incorporated into the algorithm during the iterative process. Well-known two test systems are examined through GWO and the numerical values are presented nicely. Moreover, the best value obtained by GWO is compared with other methods, and the comparison proves that the GWO is superior to other algorithms in terms of solution quality. Finally, it is concluded that the GWO provides the best solution for cost-effective dispatch and is very much useful to the system planners and this way altering their working methodologies to produce electrical vitality at affordable prices.

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